EE 316

Section 3

Pre Lab 09

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Operating Characteristics of JFETs

Introduction:

This lab introduces the Junction Field-Effect Transistor (JFET), covering n-channel and p-channel constructions, channel symmetry, and key operational parameters such as pinch-off voltage and saturation current. By examining its voltage-controlled behavior, high input impedance, and low output noise, the experiment demonstrates why JFETs excel in small-signal amplification compared to BJTs. Ultimately, it teaches how to analyze JFET biasing and operating states, establishing the foundational concepts required to understand MOSFETs in future labs.

Simulations and Results:

The JFET circuit in Figure 1 was built and simulated in LTSpice to obtain its drain and transfer characteristic curves. Using the simulated circuit in Figure 2, I_D was measured across various V_{DS} values while maintaining V_{GS} at 0 V, -0.5 V, and -1 V by adjusting V_1 and V_2 independently. These I_D values are summarized in Table 1. The resulting data was plotted to create the drain characteristic curve in Figure 3, from which V_P and V_{TH} were estimated and annotated. Next, with V_{DS} fixed at 6 V, V_{GS} was incrementally decreased until I_D reached 0 mA to generate the transfer characteristic curve in Figure 4. Finally, V_P was estimated from the transfer plot and compared against the drain characteristic value.

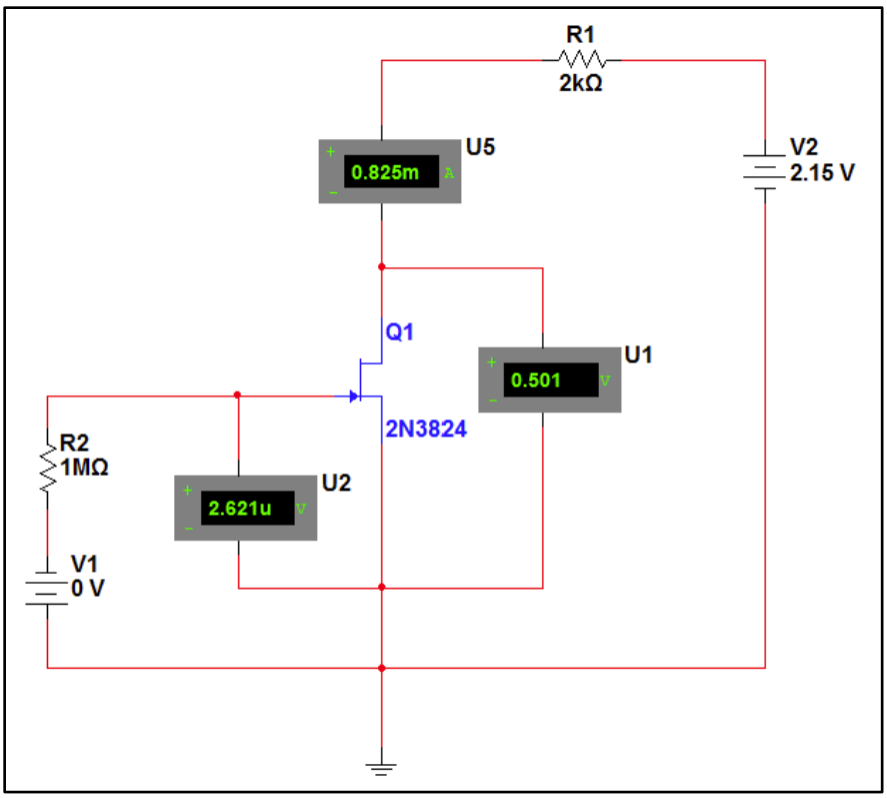


Figure #1: JFET Circuit (Lab Manual)

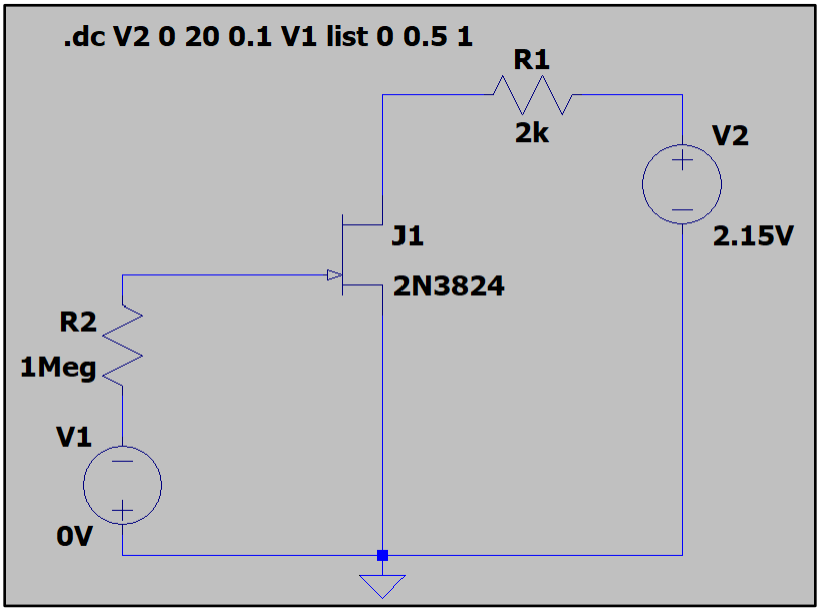


Figure #2: JFET Circuit (LTSpice)

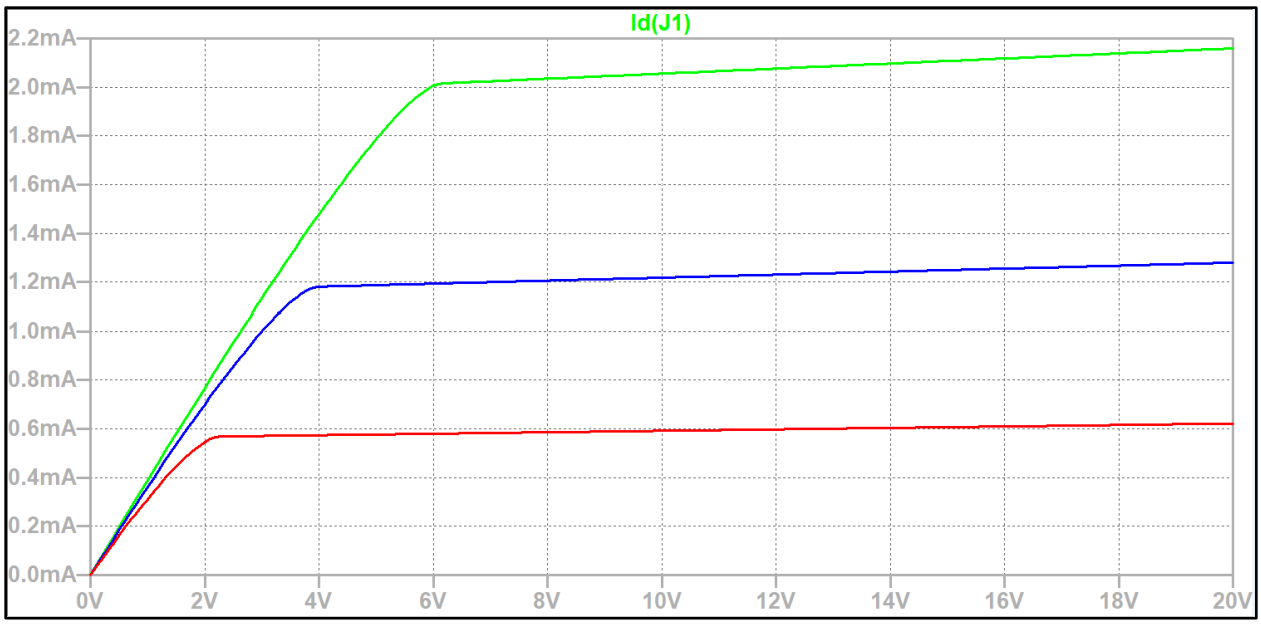


Figure #3: JFET Drain Characteristic Curves

V_P = 3.8 V

V_{TH} = -2 V

	V_{GS} = 0 V	V_{GS} = -0.5 V	V_{GS} = -1 V
V_{DS} (V)	I_D (mA)	I_D (mA)	I_D (mA)
0	0	0	0
0.5	0.19598597	0.18313265	0.16173965
1	0.38969575	0.36175252	0.31208465
2	0.76892186	0.70063752	0.54517260
4	1.4777967	1.1808369	0.57418324
8	2.0346143	1.2054729	0.58616768
12	2.076169	1.2301051	0.59815182
16	2.1177211	1.2547361	0.61013561
20	2.1592702	1.279366	0.62211906

Table #1: Drain Characteristics

	V_{DS} = 6 V
V_{GS} (V)	I_D (mA)
0	2.0565675
-0.5	1.2080352
SS-1	0.58367301
-1.5	0.18379232
-2	0.008701697

Table #2: Transfer Characteristics

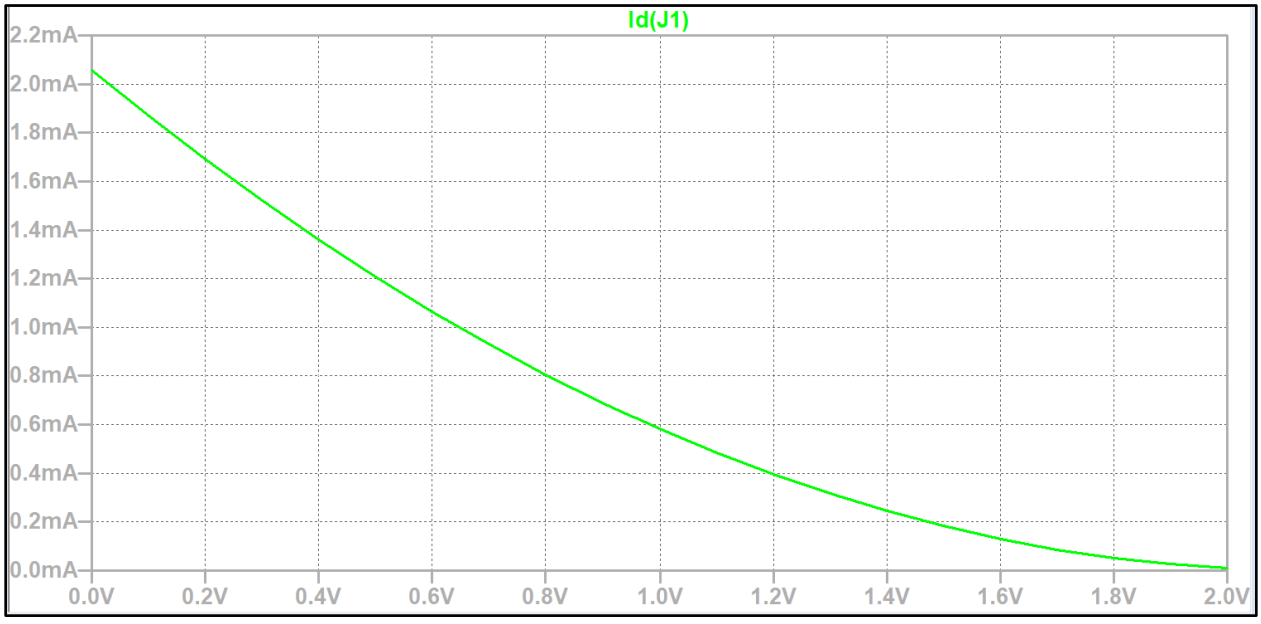


Figure #4: Transfer Characteristic Plot

V_P = 3.8 V

V_{TH} = -2 V